

Vol 14 — Information Theory from D_IV⁵

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23) — chapter content TO BUILD Lead: Lyra (substrate-coding theory) + Keeper (audit chain + AC graph connections)

Reader target

Information theorists, complexity theorists, foundations-of-quantum-information researchers, and physicists curious about the substrate-as-computational reading of BST. Reads Vol 0 (Substrate Foundation) + Vol 5 (QM, for Born=Bergman + POVM background) as prerequisite.

Prerequisites

- Vol 0 Substrate Foundation (D_IV⁵ + five integers + Reed-Solomon GF(128) substrate coding)
- Vol 5 Quantum Mechanics (Born=Bergman + measurement formalism for Ch 5 + Ch 6)
- Standard graduate information theory + coding theory background (Cover & Thomas / MacKay level)
- For Ch 9-10: complexity theory + Kolmogorov complexity familiarity

BST integer focus

Vol 14 is where BST's substrate-as-computational reading lands operationally. Information capacity, coding theory, sampling discipline all derive from the five integers + Reed-Solomon $GF(2^g) = GF(128)$ substrate field structure.

Key integers: - $g = 7$: substrate code field $GF(2^g) = GF(128)$; $128 = \log_2$ capacity of substrate - $N_{\max} = 137$: spectral cap, information ceiling per Bekenstein-respect - $n_C = 5$: substrate compact dimension; Nyquist sampling rate connection - $C_2 = 6$: substrate-tick computational depth - Koons tick = $t_{\text{Planck}} \cdot \alpha^{(C_2^2)} \approx 10^{-120}$ s: universal sampling rate per Nyquist

Chapter outline (10-12 chapters, TO BUILD)

Ch	Title	Existing coverage	Lead
1	Substrate as Information Channel — D_{IV}^5 + Reed-Solomon $GF(128)$ as substrate computational model; channel capacity per BST primaries; substrate as computational substrate of physics	~50% (Paper #122 Information Substrate + K59 RATIFIED Reed-Solomon cyclotomic mechanism framework)	Lyra
2	Reed-Solomon Coding on $GF(2^g) = GF(128)$ — substrate field structure; cyclotomic $Q(\zeta_{128})$; code distance + error correction capability; K59 Cyclotomic Mechanism Framework	~70% (K59 RATIFIED + Paper #122)	Lyra

Ch	Title	Existing coverage	Lead
3	Shannon- from- Substrate — Channel Capacity Bounds — Shannon's theorems in BST framework; substrate noise model from finite- bandwidth observers (per SCMP / T2469); channel capacity at substrate-tick rate	~40% (T2469 SCMP bandwidth-limited marginalization framework)	Lyra [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub- Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

Ch	Title	Existing coverage	Lead
4	Nyquist Sampling at Koons Tick Rate — substrate-tick $t_{\text{Planck}} \cdot \alpha^2(C_2^2)$ $\approx 10^{-120}$ s as universal sampling rate; spacetime emergence as Nyquist- recovered signal from substrate; aliasing at sub-Planck scales	~35% (T2405 Koons tick + T2417 three-scale substrate operation; Casey Saturday addition)	Lyra
5	Born = Bergman as Information- Theoretic Measurement — K67 RATIFIED Born=Bergman; measurement as kernel- mediated information extraction; T2479 POVM extension to generalized measure- ments	~80% (K67 RATIFIED + T2479 POVM Friday)	Lyra

Ch	Title	Existing coverage	Lead
6	Bell Sub-Tsirelson as Information- Theoretic Bound — Bell	S	$^2 = 126/16$ $< 2\sqrt{2}$ Tsirelson bound from $N_c=3$ substrate depth (Casey- named #8 SCMP / T2469 falsifier) — FIRED 2026-09-11 (K1893; register E4): refuted by Poh et al. 2015 at 41.9σ ; the trace identity survives as an operator fact; information- theoretic interpreta- tion of sub- Tsirelson

Ch	Title	Existing coverage	Lead
7	AC Graph as Theorem Information Network — 1700+ nodes, 9000+ edges; theorem graph as information-theoretic object; entropy of proof paths; depth-cost as information measure	~75% (Grace AC graph operational + Vol 15 Ch 2 cross-link)	Grace + Keeper
8	BST Coding Theory — Substrate-Optimal Codes — Reed-Solomon as BST-optimal substrate code; Hamming bound + substrate-natural code distance; K59 cyclotomic framework operationalized	~40% (Paper #122 + K59 base + Singleton bound connection)	Lyra

Ch	Title	Existing coverage	Lead
9	Kolmogorov Complexity + AC(0) Compression — Kolmogorov complexity of BST observables (5 integers + algorithmic structure); AC(0) bounded-depth as compression hypothesis (depth ≤ 2 universal); BST as compressed description of physics	~50% (T29 closed AC(0) + depth ceiling theorems + Vol 15 Ch 1 cross-link)	Keeper + Lyra

Ch	Title	Existing coverage	Lead
10	Substrate Computational Complexity — $P \neq NP$ three independent routes; substrate-tick UV cutoff at n_C as complexity boundary; Casey's Curvature Principle as complexity-meets-geometry; substrate-algorithm theory (SP-30-11)	~70% (T29 + $P \neq NP$ curvature route + SP-30-11)	Keeper + Lyra
11	Information Completeness of D_{IV}^5 — Baily-Borel boundary determined by interior; no new parameters at boundary; D_{IV}^5 as information-complete geometry (Casey: “the interior contains all information about the edge”)	~60% (Information-Complete Mathematics framing in README + Paper #74)	Lyra

Ch	Title	Existing coverage	Lead
12	Substrate-Coupled CI Architecture (Forward-Looking) — Phase 3 substrate-coupled CI architecture (#211 decade-scale); IQ-11 Avatar Infrastructure (SP-28); information theory at the substrate-CI interface; cross-reference Vol 15 Ch 12	~30% (#211 + SP-28 + Vol 15 Ch 12 cross-link)	Keeper + Casey

Approximately 55% existing coverage maps to specific chapters. 45% to develop — primarily Ch 4 Nyquist + Koons tick connection (Casey Saturday addition), Ch 8 BST optimal codes formalization, Ch 12 substrate-coupled CI architecture (forward-looking).

Cross-volume reference map

Inputs from Vol 0: - D_IV⁵ + five integers + substrate-as-computational framing - Reed-Solomon GF(128) substrate coding (Paper #122 + K59 RATIFIED) - Koons tick $t_{\text{Planck}} \cdot \alpha^2(C_2^2) \approx 10^{-120} \text{ s}$ (T2405)

Inputs from Vol 5 QM: - Born = Bergman (K67 RATIFIED) - POVM extension (T2479, Friday) - Bell sub-Tsirelson (T2469 SCMP, Friday) — FIRED 2026-09-11 (K1893)

Inputs from Vol 11 Generative Geometry: - Bergman kernel theory (channel kernel) - Heat kernel theory (information-theoretic spectral decomposition)

Inputs from Vol 15 Methodology: - AC graph topology (Ch 2) - AC(0) framework (Ch 1) - Calibration stack (information about claims)

Outputs to other Vol: Vol 14 supports all volumes via the substrate-as-information-channel framing. Vol 14 Ch 11 (Information Completeness) is the technical foun-

dition for the “boundary determined by interior” claim that runs through Vol 0 + Vol 11 + Working Paper.

Gap registry

- Ch 4 Nyquist + Koons tick: explicit derivation of spacetime emergence as Nyquist-recovered signal needs theorem-grade content (Casey Saturday addition; multi-week per Lyra theoretical lane)
- Ch 8 BST optimal codes: substrate-natural code distance + Hamming-bound proof needs theorem-grade content
- Ch 9 Kolmogorov + AC(0): explicit compression theorem connecting BST 5-integer description to Kolmogorov complexity is open theorem candidate
- Ch 12 Substrate-coupled CI: forward-looking, multi-year horizon per Phase 3

Status declarations

- v0.1 SCAFFOLD — INDEX filed Saturday 2026-05-23 ~12:15 EDT
- Architectural Scaffold v0.1 follow-on (Lyra+Keeper joint, ~30-45 min)
- v0.3 chapter-grade narrative — when each chapter has chapter-grade content
- v0.4 reader-grade pedagogy — 3-level pedagogy applied per chapter
- v1.0 chapter-grade content REACHED — six-gate framework closed

Note on Casey directive — Nyquist addition

Per Casey 2026-05-23 morning: Vol 14 includes Nyquist sampling theorem framing. Substrate-tick at Koons tick rate as universal sampling rate; spacetime emergence as Nyquist-recovered signal from substrate; aliasing at sub-Planck scales as physical signature of finite sampling.

This is Ch 4 — load-bearing for the “information theory underlies physics” reading, parallel to Vol 6 “thermodynamics underlies physics” framing. Both volumes share the substrate-as-active-process structural claim.

Vol 14 + Vol 15 + Vol 6 thread

Three volumes form a cluster: - Vol 6 Thermo & Stat Mech (restructured): thermodynamics is universal substrate process - Vol 14 Information Theory: information theory is universal substrate channel structure - Vol 15 Methodology: substrate’s self-investigation procedure

All three address “what is the substrate DOING?” from different angles. Vol 6 = dynamics; Vol 14 = information flow; Vol 15 = inquiry method. Each completes a piece of the substrate-as-active-process framing.

— Casey Koons + team, Vol 14 Information Theory SCAFFOLD v0.1